Microeconomics

Course Code : Hum 2101

Lecture 2: Circular Flow of Economics, Production Possibility frontier

Today we will learn -

- Circular Flow of Economics
- How to read Graphs
- Production Possibility frontier
- Opportunity Cost

Circular Flow of Economics

- **Two types of decision makers—firms, and households**

Firms produce goods and services using inputs, such as labor, land, and capital (buildings and machines). These inputs are called the factors of production.

Households own the factors of production and consume all the goods and services that the firms produce.

Circular Flow of Economics

- **Households and firms interact in two types of markets.**

Markets for goods and services

- households are buyers
- firms are sellers
- Households buy the output of goods and services that firms produce

Markets for the factors of production

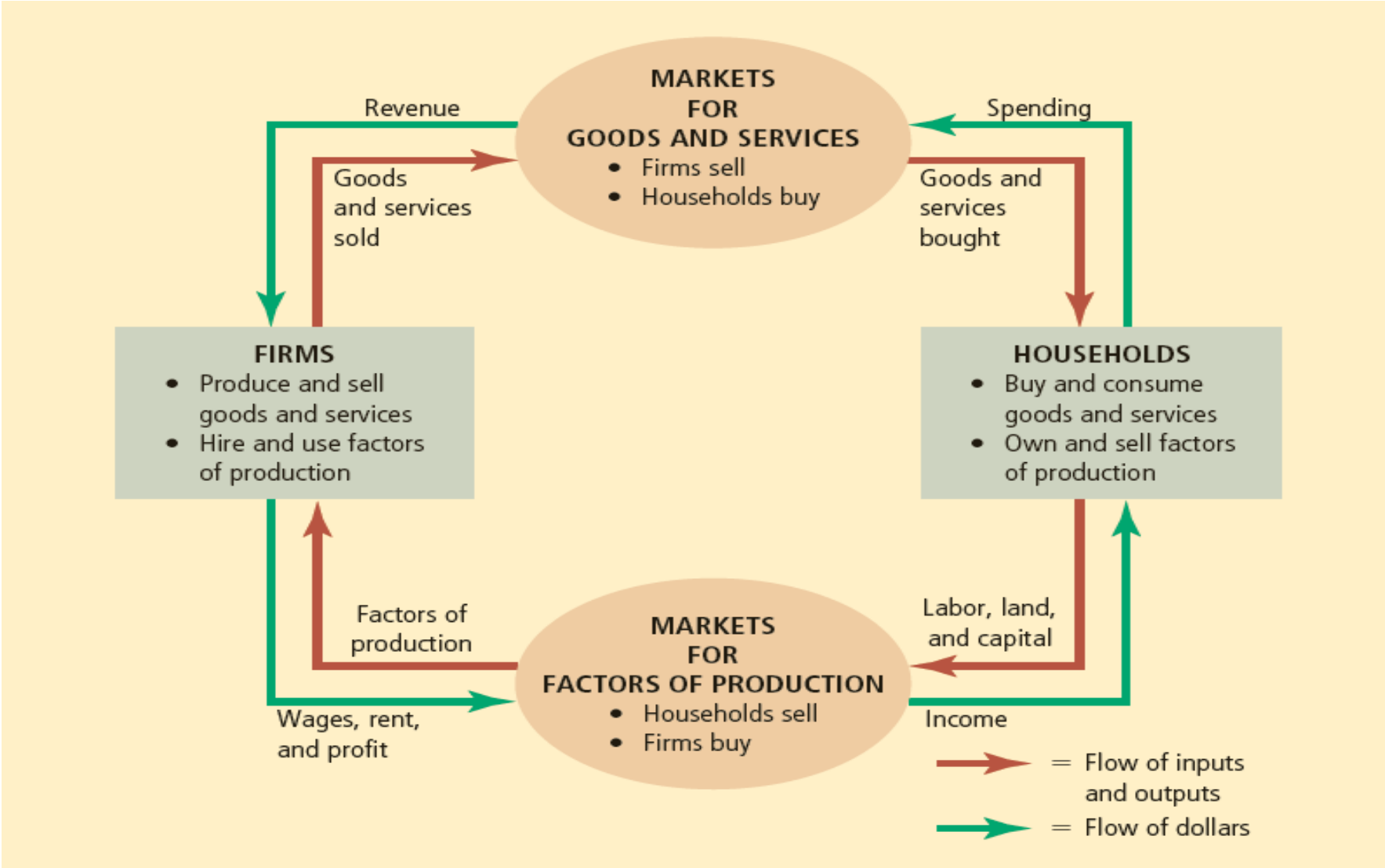
- households are sellers
- firms are buyers
- households provide the inputs that firms use to produce goods and services

Circular Flow of Economics

Two loops of the circular-flow diagram are distinct but related –

- The inner loop represents the flows of inputs and outputs.
- The outer loop of the diagram represents the corresponding flow of dollars.

Circular Flow of Economics



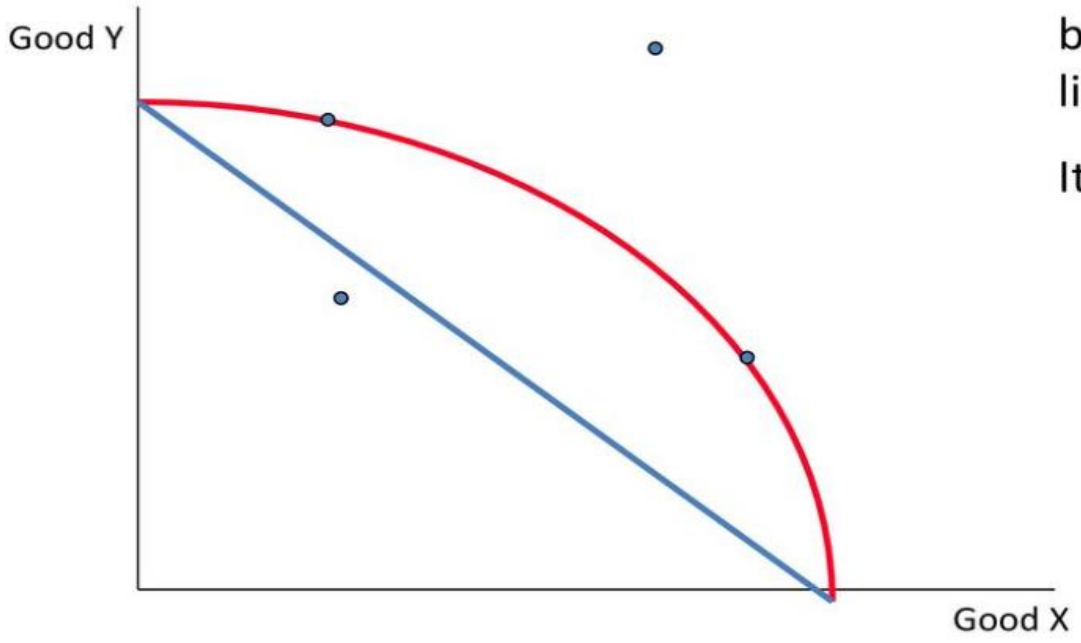
The Production Possibilities Frontier

The quantities of goods and services that we can produce are limited both by our available resources and by technology. And if we want to increase our production of one good, we must decrease our production of something else—we face a tradeoff.

The production possibilities frontier (PPF) is the boundary between those combinations of goods and services that can be produced and those that cannot.

The Production Possibilities Frontier

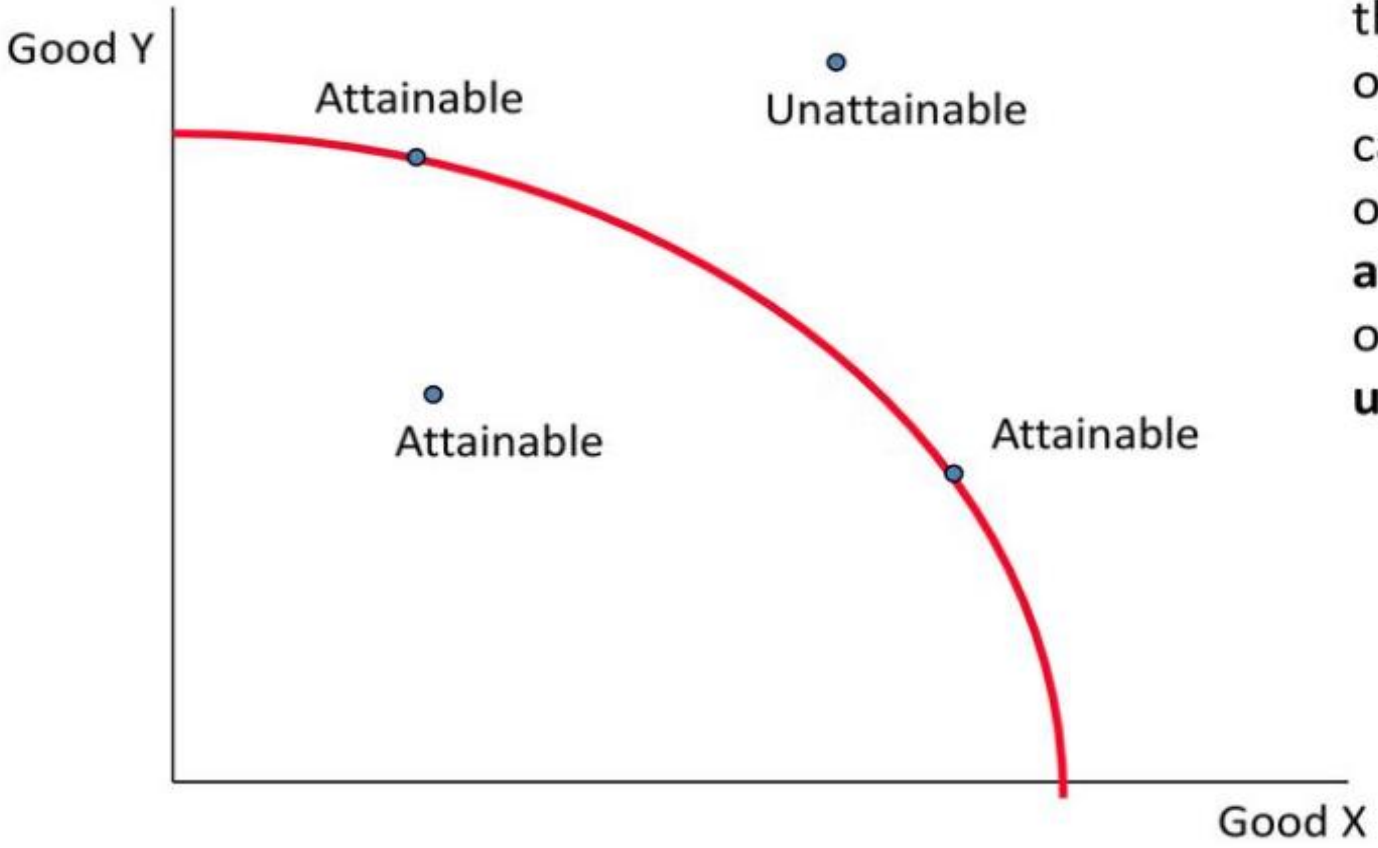
A Typical PPF Picture



The PPF is typically bowed-out or linear.

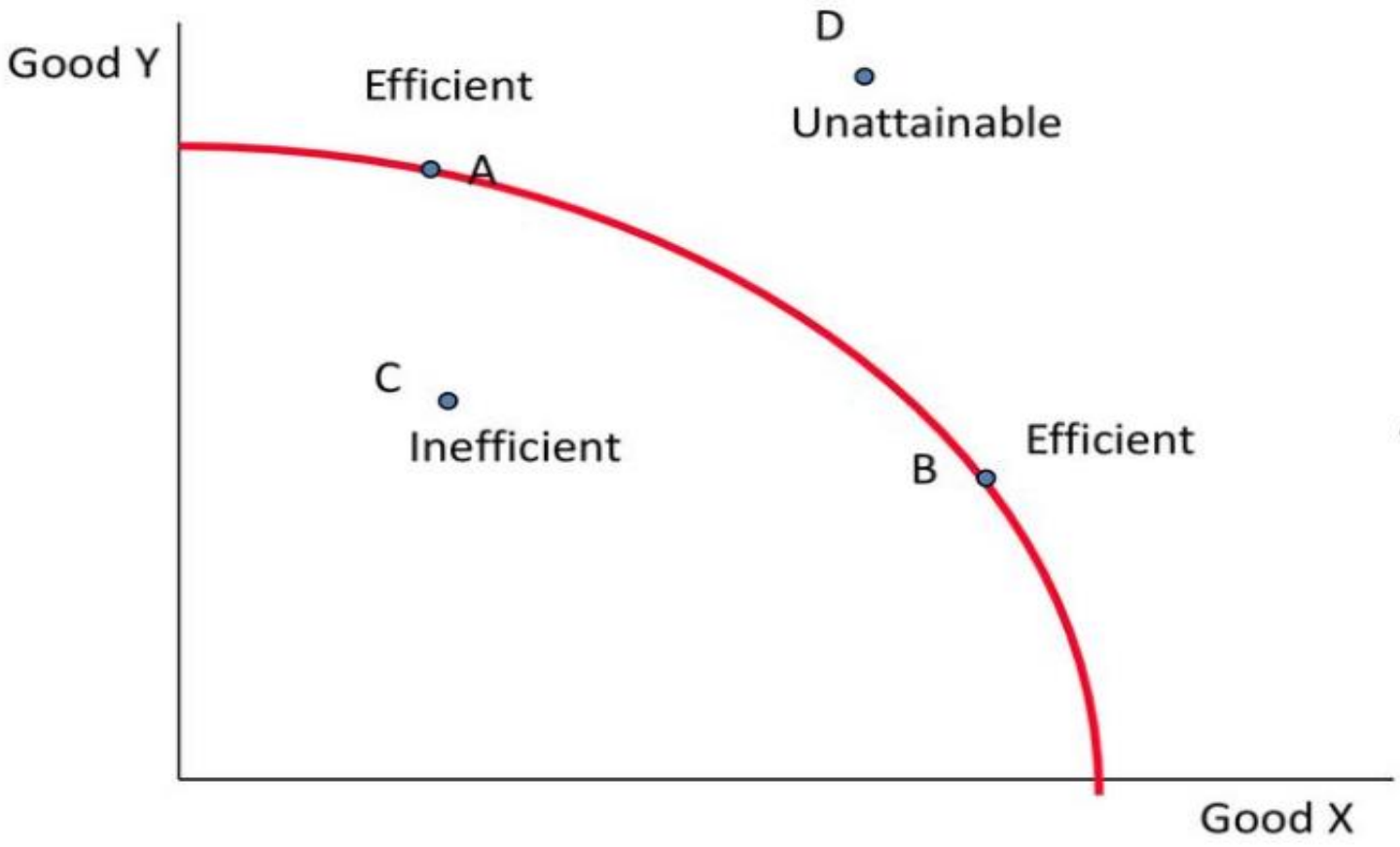
It is not bowed-in

The Production Possibilities Frontier



Since the PPF shows the limits to outputs of our economy, we can see directly, which outputs levels are **attainable** and which ones are **unattainable**.

The Production Possibilities Frontier



- According to the PPF points A and B - appearing on the curve – represent the most efficient use of resources by the economy.
- Point C represents an inefficient use of resources, while point D represents the goals that the economy can not attain with its present levels of resources.

The Production Possibilities Frontier

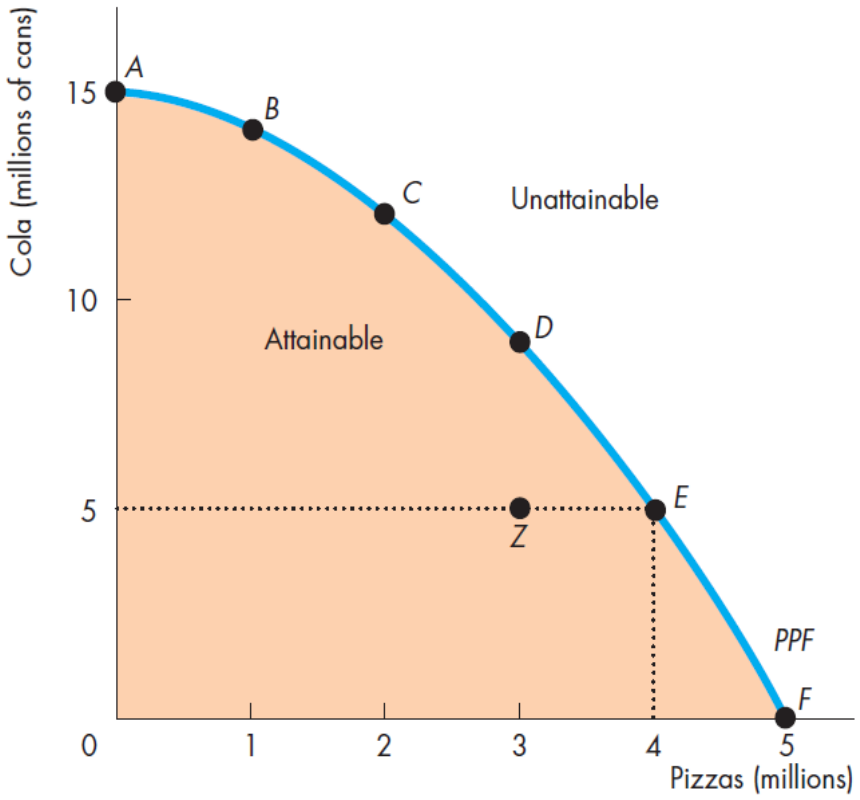
To illustrate the PPF,

- we focus on two goods at a time, and
- hold the quantities produced of all the other goods and services constant.

Let's look at the production possibilities frontier for cola and pizza, which represent any pair of goods or services.

The Production Possibilities Frontier

Possibility	Pizzas (millions)		Cola (millions of cans)
A	0	and	15
B	1	and	14
C	2	and	12
D	3	and	9
E	4	and	5
F	5	and	0

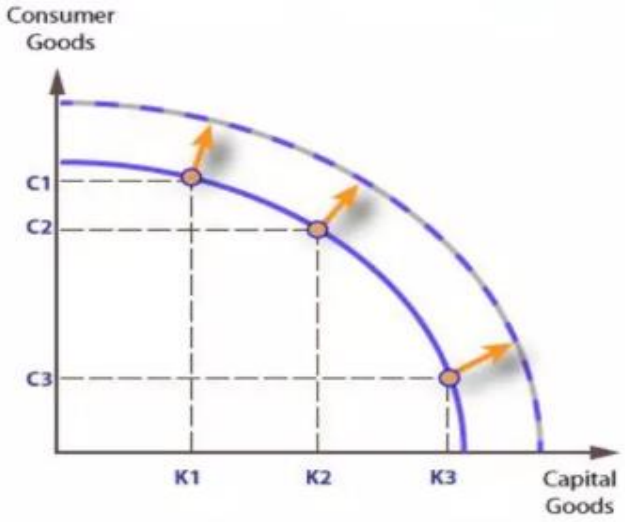


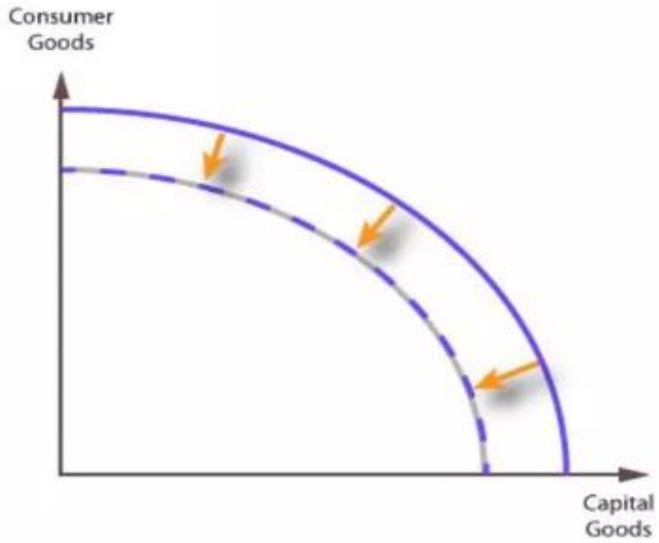
The Production Possibilities Frontier

Some observations from the graph:

- ➢The x-axis shows the quantity of pizzas produced
- ➢The y-axis shows the quantity of cola produced.
- The PPF separates the *attainable* from the unattainable.
- Production is possible at any point inside the orange area or on the frontier. Points outside the frontier are *unattainable*.
- Points inside the frontier, such as point Z, are *inefficient.*

The Production Possibilities Frontier





Outward Shift

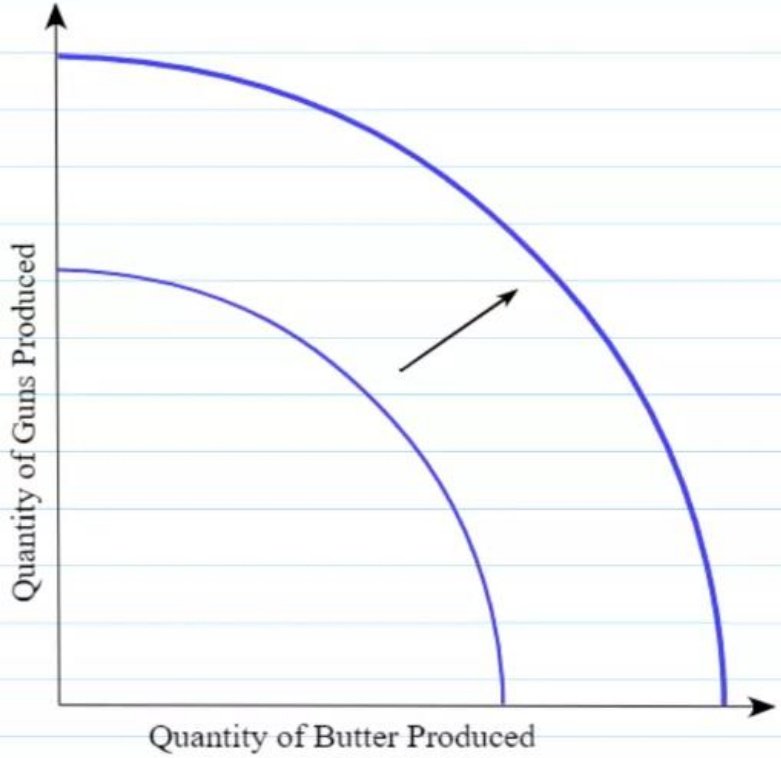
- New Technology
- New production method
- New Raw materials
- Increase in labor force

Inward Shift

- Resources run out
- Natural disaster
- Failure to invest
- Erosion of infrastructure

The Production Possibilities Frontier

What happens when technological advancement occurs?



- Shifts the PPFC rightward
- Less costly inputs are needed

Production Efficiency

We achieve **production efficiency** if we produce goods and services at the lowest possible cost.

Production is inefficient inside the PPF because resources are either ***unused*** or ***misallocated*** or ***both***.

Tradeoff Along the PPF

- Every choice along the PPF involves a tradeoff.
- This boundary is the real world's production possibilities frontier, and it defines the tradeoffs that we must make.
- On our real-world PPF, we can produce more of any one good or service only if we produce less of some other goods or services.
- All tradeoffs involve a cost—an opportunity cost.

Warm up question:

Tell me about a choice that you made yesterday. What was the opportunity cost of that choice?

Opportunity Cost

In a world of scarcity, choosing one thing means giving up something else. The opportunity cost of a decision is the value of the good or service forgone.

Opportunity Cost

When economists refer to the “opportunity cost” of a resource, they mean the value of the next-highest-valued alternative use of that resource.

If, for example, you spend time and money going to a movie, you cannot spend that time at home reading a book, and you can’t spend the money on something else. If your next-best alternative to seeing the movie is reading the book, then the opportunity cost of seeing the movie is the money spent plus the pleasure you forgo by not reading the book….